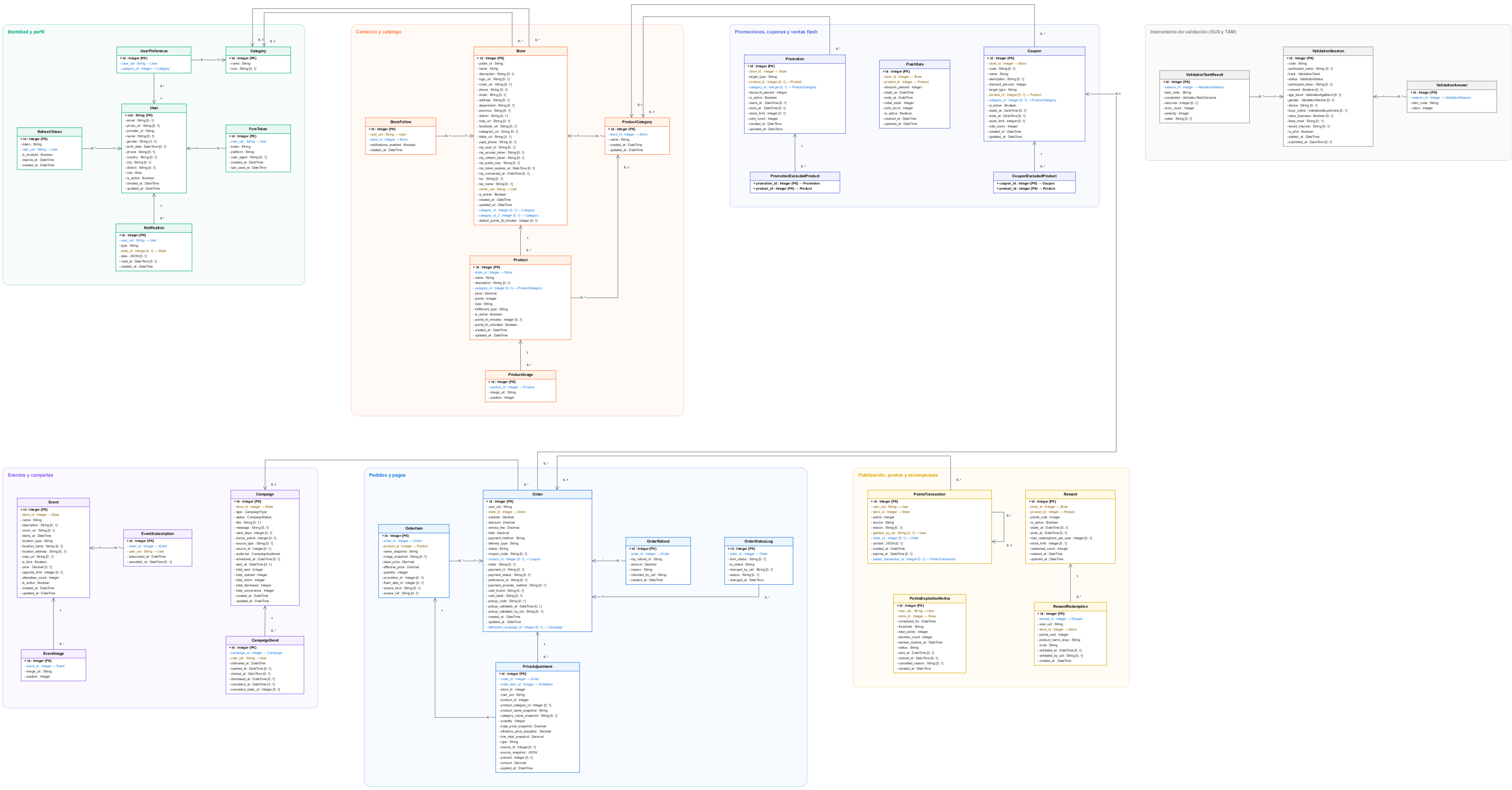
# Diagrama de clases completo del modelo de dominio: 33 clases, 346 atributos y 56 asociaciones

Cada bloque es un paquete y conserva la disposición de su vista de detalle. + marca la clave primaria {PK}; [0..1] indica que el atributo admite nulo y [*] que es una lista. En azul, las claves foráneas cuya asociación se dibuja, con su multiplicidad en ambos extremos.
En ocre, las 25 claves foráneas que apuntan a User, Store o Product desde otro paquete: casi todas las clases referencian a esos tres concentradores, y se indican en el atributo en lugar de dibujarse para no saturar el diagrama.


Identidad y perfil
UserPreference
+ id : Integer {PK}
- user_uid : String → User
- category_id : Integer → Category
Category
+ id : Integer {PK}
- name : String
- icon : String [0..1]
User
+ uid : String {PK}
- email : String [0..1]
- photo_url : String [0..1]
- provider_id : String
- name : String [0..1]
- gender : String [0..1]
- birth_date : DateTime [0..1]
- phone : String [0..1]
- country : String [0..1]
- city : String [0..1]
- district : String [0..1]
- role : Role
- is_active : Boolean
- created_at : DateTime
- updated_at : DateTime
RefreshToken
+ id : Integer {PK}
- token : String
- user_uid : String → User
- is_revoked : Boolean
- expires_at : DateTime
- created_at : DateTime
FcmToken
+ id : Integer {PK}
- user_uid : String → User
- token : String
- platform : String
- user_agent : String [0..1]
- created_at : DateTime
- last_used_at : DateTime
Notification
+ id : Integer {PK}
- user_uid : String → User
- type : String
- store_id : Integer [0..1] → Store
- data : JSON [0..1]
- read_at : DateTime [0..1]
- created_at : DateTime
Comercio y catálogo
Store
+ id : Integer {PK}
- public_id : String
- name : String
- description : String [0..1]
- logo_url : String [0..1]
- cover_url : String [0..1]
- phone : String [0..1]
- email : String [0..1]
- address : String [0..1]
- department : String [0..1]
- province : String [0..1]
- district : String [0..1]
- map_url : String [0..1]
- facebook_url : String [0..1]
- instagram_url : String [0..1]
- tiktok_url : String [0..1]
- yape_phone : String [0..1]
- mp_user_id : String [0..1]
- mp_access_token : String [0..1]
- mp_refresh_token : String [0..1]
- mp_public_key : String [0..1]
- mp_token_expires_at : DateTime [0..1]
- mp_connected_at : DateTime [0..1]
- ruc : String [0..1]
- tax_name : String [0..1]
- owner_uid : String → User
- is_active : Boolean
- created_at : DateTime
- updated_at : DateTime
- category_id : Integer [0..1] → Category
- category_id_2 : Integer [0..1] → Category
- default_points_ttl_minutes : Integer [0..1]
StoreFollow
+ id : Integer {PK}
- user_uid : String → User
- store_id : Integer → Store
- notifications_enabled : Boolean
- created_at : DateTime
ProductCategory
+ id : Integer {PK}
- store_id : Integer → Store
- name : String
- created_at : DateTime
- updated_at : DateTime
Product
+ id : Integer {PK}
- store_id : Integer → Store
- name : String
- description : String [0..1]
- category_id : Integer [0..1] → ProductCategory
- price : Decimal
- points : Integer
- type : String
- fulfillment_type : String
- is_active : Boolean
- points_ttl_minutes : Integer [0..1]
- points_ttl_unlimited : Boolean
- created_at : DateTime
- updated_at : DateTime
ProductImage
+ id : Integer {PK}
- product_id : Integer → Product
- image_url : String
- position : Integer
Promociones, cupones y ventas flash
Promotion
+ id : Integer {PK}
- store_id : Integer → Store
- target_type : String
- product_id : Integer [0..1] → Product
- category_id : Integer [0..1] → ProductCategory
- discount_percent : Integer
- is_active : Boolean
- starts_at : DateTime [0..1]
- ends_at : DateTime [0..1]
- stock_limit : Integer [0..1]
- sold_count : Integer
- created_at : DateTime
- updated_at : DateTime
FlashSale
+ id : Integer {PK}
- store_id : Integer → Store
- product_id : Integer → Product
- discount_percent : Integer
- starts_at : DateTime
- ends_at : DateTime
- initial_stock : Integer
- sold_count : Integer
- is_active : Boolean
- created_at : DateTime
- updated_at : DateTime
Coupon
+ id : Integer {PK}
- store_id : Integer → Store
- code : String [0..1]
- name : String
- description : String [0..1]
- discount_percent : Integer
- target_type : String
- product_id : Integer [0..1] → Product
- category_id : Integer [0..1] → ProductCategory
- is_active : Boolean
- starts_at : DateTime [0..1]
- ends_at : DateTime [0..1]
- stock_limit : Integer [0..1]
- sold_count : Integer
- created_at : DateTime
- updated_at : DateTime
PromotionExcludedProduct
+ promotion_id : Integer {PK} → Promotion
+ product_id : Integer {PK} → Product
CouponExcludedProduct
+ coupon_id : Integer {PK} → Coupon
+ product_id : Integer {PK} → Product
Instrumento de validación (SUS y TAM)
ValidationTaskResult
+ id : Integer {PK}
- session_id : Integer → ValidationSession
- task_code : String
- completed : ValidationTaskOutcome
- seconds : Integer [0..1]
- error_count : Integer
- severity : Integer
- notes : String [0..1]
ValidationSession
+ id : Integer {PK}
- code : String
- participant_name : String [0..1]
- track : ValidationTrack
- status : ValidationStatus
- participant_token : String [0..1]
- consent : Boolean [0..1]
- age_band : ValidationAgeBand [0..1]
- gender : ValidationGender [0..1]
- device : String [0..1]
- buys_online : ValidationBuysOnline [0..1]
- owns_business : Boolean [0..1]
- liked_most : String [0..1]
- would_improve : String [0..1]
- is_pilot : Boolean
- started_at : DateTime
- submitted_at : DateTime [0..1]
ValidationAnswer
+ id : Integer {PK}
- session_id : Integer → ValidationSession
- item_code : String
- value : Integer
Eventos y campañas
Event
+ id : Integer {PK}
- store_id : Integer → Store
- name : String
- description : String [0..1]
- cover_url : String [0..1]
- starts_at : DateTime
- location_type : String
- location_name : String [0..1]
- location_address : String [0..1]
- map_url : String [0..1]
- is_free : Boolean
- price : Decimal [0..1]
- capacity_limit : Integer [0..1]
- attendees_count : Integer
- is_active : Boolean
- created_at : DateTime
- updated_at : DateTime
EventSubscription
+ id : Integer {PK}
- event_id : Integer → Event
- user_uid : String → User
- subscribed_at : DateTime
- cancelled_at : DateTime [0..1]
EventImage
+ id : Integer {PK}
- event_id : Integer → Event
- image_url : String
- position : Integer
Campaign
+ id : Integer {PK}
- store_id : Integer → Store
- type : CampaignType
- status : CampaignStatus
- title : String [0..1]
- message : String [0..1]
- valid_days : Integer [0..1]
- bonus_points : Integer [0..1]
- source_type : String [0..1]
- source_id : Integer [0..1]
- audience : CampaignAudience
- scheduled_at : DateTime [0..1]
- sent_at : DateTime [0..1]
- total_sent : Integer
- total_opened : Integer
- total_clicks : Integer
- total_dismissed : Integer
- total_conversions : Integer
- created_at : DateTime
- updated_at : DateTime
CampaignSend
+ id : Integer {PK}
- campaign_id : Integer → Campaign
- user_uid : String → User
- delivered_at : DateTime
- opened_at : DateTime [0..1]
- clicked_at : DateTime [0..1]
- dismissed_at : DateTime [0..1]
- converted_at : DateTime [0..1]
- converted_order_id : Integer [0..1]
Pedidos y pagos
Order
+ id : Integer {PK}
- user_uid : String
- store_id : Integer → Store
- subtotal : Decimal
- discount : Decimal
- service_fee : Decimal
- total : Decimal
- payment_method : String
- delivery_type : String
- status : String
- coupon_code : String [0..1]
- coupon_id : Integer [0..1] → Coupon
- notes : String [0..1]
- payment_id : String [0..1]
- payment_status : String [0..1]
- preference_id : String [0..1]
- payment_provider_method : String [0..1]
- card_brand : String [0..1]
- card_last4 : String [0..1]
- pickup_code : String [0..1]
- pickup_validated_at : DateTime [0..1]
- pickup_validated_by_uid : String [0..1]
- created_at : DateTime
- updated_at : DateTime
- attributed_campaign_id : Integer [0..1] → Campaign
OrderItem
+ id : Integer {PK}
- order_id : Integer → Order
- product_id : Integer → Product
- name_snapshot : String
- image_snapshot : String [0..1]
- base_price : Decimal
- effective_price : Decimal
- quantity : Integer
- promotion_id : Integer [0..1]
- flash_sale_id : Integer [0..1]
- source_kind : String [0..1]
- source_ref : String [0..1]
OrderRefund
+ id : Integer {PK}
- order_id : Integer → Order
- mp_refund_id : String
- amount : Decimal
- reason : String
- refunded_by_uid : String
- created_at : DateTime
OrderStatusLog
+ id : Integer {PK}
- order_id : Integer → Order
- from_status : String [0..1]
- to_status : String
- changed_by_uid : String [0..1]
- reason : String [0..1]
- changed_at : DateTime
PriceAdjustment
+ id : Integer {PK}
- order_id : Integer → Order
- order_item_id : Integer → OrderItem
- store_id : Integer
- user_uid : String
- product_id : Integer
- product_category_id : Integer [0..1]
- product_name_snapshot : String
- category_name_snapshot : String [0..1]
- quantity : Integer
- base_price_snapshot : Decimal
- effective_price_snapshot : Decimal
- line_total_snapshot : Decimal
- type : String
- source_id : Integer [0..1]
- source_snapshot : JSON
- percent : Integer [0..1]
- amount : Decimal
- applied_at : DateTime
Fidelización: puntos y recompensas
PointsTransaction
+ id : Integer {PK}
- user_uid : String → User
- store_id : Integer → Store
- points : Integer
- source : String
- reason : String [0..1]
- granted_by_uid : String [0..1] → User
- order_id : Integer [0..1] → Order
- context : JSON [0..1]
- created_at : DateTime
- expires_at : DateTime [0..1]
- parent_transaction_id : Integer [0..1] → PointsTransaction
Reward
+ id : Integer {PK}
- store_id : Integer → Store
- product_id : Integer → Product
- points_cost : Integer
- is_active : Boolean
- starts_at : DateTime [0..1]
- ends_at : DateTime [0..1]
- max_redemptions_per_user : Integer [0..1]
- stock_limit : Integer [0..1]
- redeemed_count : Integer
- created_at : DateTime
- updated_at : DateTime
PointsExpirationNotice
+ id : Integer {PK}
- user_uid : String → User
- store_id : Integer → Store
- scheduled_for : DateTime
- threshold : String
- total_points : Integer
- batches_count : Integer
- earliest_expires_at : DateTime
- status : String
- sent_at : DateTime [0..1]
- clicked_at : DateTime [0..1]
- cancelled_reason : String [0..1]
- created_at : DateTime
RewardRedemption
+ id : Integer {PK}
- reward_id : Integer → Reward
- user_uid : String
- store_id : Integer → Store
- points_cost : Integer
- product_name_snap : String
- code : String
- validated_at : DateTime [0..1]
- validated_by_uid : String [0..1]
- created_at : DateTime